

Md. Aman Uddin Siyam

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SUMMARY

BSc in Computer Science candidate (all coursework completed, available immediately) and ML Researcher with a proven track record of architecting end-to-end deep learning pipelines and publishing peer-reviewed benchmarks. Experienced in computer vision, biomedical signal processing, and time-series forecasting, with a strong focus on MLOps, edge-device optimization, and explainable AI. Seeking a **Machine Learning Engineering** or **MLOps Internship** to build production-ready, research-driven AI solutions.

KEY SKILLS

Programming	Python, NumPy, Pandas, Matplotlib, scikit-learn
Deep Learning & CV	TensorFlow/Keras, PyTorch, OpenCV, Transfer Learning, Grad-CAM
Signal & Time-Series	Biomedical Signal Processing, Feature Extraction, ARIMA, Prophet, LSTM, PCA, Survival Analysis
MLOps & Deployment	Docker, FastAPI/Flask, REST APIs, Weights & Biases (W&B), Reproducible Pipelines, Experiment Logging
Data & Research	Exploratory Data Analysis (EDA), Feature Engineering, Cross-Validation, Experimental Design, SHAP
Tools & Platforms	Git/GitHub, Jupyter, Google Colab, Kaggle

SELECTED PROJECTS

BAT-Track: Energy-Efficient Visual Tracking for UAVs

- Developed a lightweight computer vision pipeline for real-time object tracking on resource-constrained UAVs, achieving **39.3 FPS** during rapid aerial maneuvers.
- Implemented and optimized state-of-the-art trackers (**OSTrack**, **SiamAPN++**) using an adaptive switching mechanism, reducing power consumption by **60.5%** while maintaining a **64.2%** tracking accuracy (IoU).

 [GitHub: BAT-Track](#) |  [Live Tracking](#)

Edge-ViT-FSL: Resource-Aware Few-Shot Learning for Cross-Domain Plant Disease Diagnosis on IoT Devices

- Architected a CPU-optimized, cross-domain crop disease diagnosis system to identify plant pathologies in real-world environments, operating under **10MB** and achieving **22.13 FPS**.
- Integrated a lightweight **MobileViT-XS** backbone with a custom Saliency-Guided Attention Module (SGAM), achieving **73.93%** cross-domain accuracy using only **5** training images per class.

 [GitHub: Edge-vit-disease-detect](#) |  [Live App](#)

Financial RAG Analyst

- Engineered a production-grade RAG pipeline using **LlamaIndex**, **Qdrant**, and **Llama 3 (via Groq)** to ingest, compare, and analyze multi-hundred-page SEC 10-K filings for Apple, Microsoft, and Google.
- Architected an **AI Observability** dashboard to monitor system latency, flag hallucinations, and capture **RLHF** user feedback via a secure logging system.
- Integrated **LlamaParse** for complex financial table extraction, containerized the application via **Docker**, and automated deployment to Streamlit Cloud using **GitHub Actions CI/CD**.

 [GitHub: Financial-RAG-Analyst](#) |  [Live App](#)

Clinical vs. Ultrasound for PCOS Detection: A Calibrated, Real-Time Benchmark

- Engineered a unified, leakage-safe machine learning benchmark to evaluate PCOS detection, comparing a clinical soft-voting ensemble against a **ConvNeXt-Small** deep learning image pipeline.
- Trained and evaluated models under an identical 5-fold out-of-fold protocol using **541** clinical records and **3,200** ultrasound images.
- Proved that the explicitly calibrated clinical ensemble (**XGBoost+LightGBM+CatBoost**) substantially outperformed the image model, achieving an AUROC of **0.976** (vs. 0.814) and 2.6x higher batch throughput.

 [GitHub: Ultrasound-PCOS](#) |  [GitHub: Tabuler-PCOS](#)

EXPERIENCE

Co-Founder & Researcher [CollabCircle](#): Lead collaborative ML/DL research projects, coordinate experiments, mentor contributors, and manage roadmaps.

RESEARCH & PUBLICATIONS (SELECTED)

- Voice-Based Parkinson's Disease Detection using SVM & XGBoost. ([Paper Link](#))
- Clinical vs. Ultrasound for PCOS Detection: A Calibrated, Leakage-Safe, Real-Time Benchmark (Accepted at ICCIT).
- Reproducible Machine Learning Pipeline for Predicting Cosmetic Chemical Carcinogenicity and Reproductive Toxicity Using Molecular Descriptors and Fingerprints (Accepted at ICCIT).

CERTIFICATIONS & TRAINING

- **Machine Learning with Python** – EDGE Bangladesh [\[Credential\]](#)
- **Introduction to Generative AI** – Google Cloud [\[Credential\]](#)
- **Advanced Machine Learning Micro-Courses** (CV, DL, Intermediate ML) – Kaggle [\[Credential\]](#)

HACKATHONS & COMPETITIONS

- **Solvio AI Hackathon** – Competitor (2025)
- **ICPC Dhaka Regional Preliminary Contest** – Competitor (2024)

EDUCATION

BSc in Computer Science & Engineering	2022 – Present
International University of Business Agriculture and Technology (IUBAT)	CGPA: 3.24

HOBBIES

- Playing cricket
- Fishing

Last updated: February 24, 2026